Curriculum Vitae

Maarten R. Dobbelaere
Scientist, Laboratory for Artificial Chemical Intelligence (LIAC), EPFL, Switzerland
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16	**6**	**557**	**12**	**5**
peer-reviewed papers	as first author	citations	h-index	PhD candidates co-supervised

Citations and h-index from Web of Science (ResearcherID ABD-4954-2021), 12 October 2026.

Research interests

Artificial intelligence and computational chemistry for chemicals and materials that are safe and sustainable by design: predicting how a molecule or material will perform, how it can be made and how it will degrade before it reaches the lab. I combine machine learning with organic synthesis, reaction engineering and green chemistry, from reaction planning and solvent selection to reactors.

Academic appointments

Scientist (Collaborateur scientifique) Aug 2026 – present
Laboratory for Artificial Chemical Intelligence (LIAC), EPFL, Switzerland
AI for chemical synthesis planning.

Visiting scientist Feb 2026 – Jun 2026
Laboratory for Artificial Chemical Intelligence (LIAC), EPFL, Switzerland
Research stay supported by the Research Foundation – Flanders (FWO).

FWO Junior Postdoctoral Fellow Oct 2025 – Jul 2026
Ghent University and Flemish Institute for Biotechnology (VIB), Belgium
Design of agrochemicals using a self-driving, high-throughput laboratory.

Postdoctoral researcher Jul 2025 – Sep 2025
Laboratory for Chemical Technology, Ghent University, Belgium
Modeling of the biodegradability of pharmaceuticals; TransPharm project.

Visiting scientist Jun 2025
Green Chemistry Centre of Excellence, University of York, UK
Machine learning for greener solvent selection; TransPharm project.

Education

PhD in Chemical Engineering Sep 2020 – May 2025
Laboratory for Chemical Technology, Ghent University, Belgium
Thesis: *Molecular Machine Learning for Chemical Reaction Engineering*.
Supervisors: Prof. Kevin M. Van Geem and Prof. Christian V. Stevens. Personal FWO fellowship.

MSc in Chemical Engineering, magna cum laude Sep 2018 – Jun 2020
Ghent University, Belgium, with an exchange semester at NTNU, Norway
Thesis: *Machine Learning Based Prediction of Thermochemical Properties*.

BSc in Chemical Engineering and Materials Science, cum laude Sep 2015 – Jun 2018
Ghent University, Belgium
Thesis: *Design of Sustainable Catalysts for Transesterification*.

Fellowships and grants

FWO Junior Postdoctoral Fellowship. Three-year award, approximately €300,000.	2025
FWO PhD Fellowship, strategic basic research. Four-year award, approximately €250,000.	2021
FWO grant for a long stay abroad, €6,864	2026
UGent CWO research-stay grant, €5,500	2026
UGent CWO travel grants, €1,000 each	2023, 2024
Erasmus+ mobility grant, €1,650	2018

Honours and awards

Finalist, Flemish PhD Cup (science communication competition)	2026
Poster prize, AIchemy Inaugural Annual Conference, Imperial College London	2025
High-impact research paper award, Chinese Academy of Engineering	2023
Dow Benelux Innovative Solutions Award for best thesis in chemistry and engineering	2020
Finalist, Jozef Plateau Prize for the best engineering thesis at Ghent University	2020

Selected publications

16 peer-reviewed publications (6 as first author) and 2 preprints. The complete list is provided as a separate document.

Maarten R. Dobbelaere, Helen F. Sneddon. The Stubborn Persistence of Toxic Solvents in Chemical Synthesis. *Angewandte Chemie International Edition* **2026**, e1677569.

Maarten R. Dobbelaere, István Lengyel, Christian V. Stevens, Kevin M. Van Geem. Geometric deep learning for molecular property predictions with chemical accuracy across chemical space. *Journal of Cheminformatics* **2024**, 16, 99.

Maarten R. Dobbelaere, István Lengyel, Christian V. Stevens, Kevin M. Van Geem. Rxn-INSIGHT: fast chemical reaction analysis using bond-electron matrices. *Journal of Cheminformatics* **2024**, 16, 37.

Maarten R. Dobbelaere, Yannick Ureel, Florence H. Vermeire, Lowie Tomme, Christian V. Stevens, Kevin M. Van Geem. Machine Learning for Physicochemical Property Prediction of Complex Hydrocarbon Mixtures. *Industrial & Engineering Chemistry Research* **2022**, 61(24), 8581–8594.

Maarten R. Dobbelaere, Pieter P. Plehiers, Ruben Van de Vijver, Christian V. Stevens, Kevin M. Van Geem. Machine Learning in Chemical Engineering: Strengths, Weaknesses, Opportunities, and Threats. *Engineering* **2021**, 7(9), 1201–1211.

Maarten R. Dobbelaere, Pieter P. Plehiers, Ruben Van de Vijver, Christian V. Stevens, Kevin M. Van Geem. Learning Molecular Representations for Thermochemistry Prediction of Cyclic Hydrocarbons and Oxygenates. *Journal of Physical Chemistry A* **2021**, 125(23), 5166–5179.

Research software and data

Rxn-INSIGHT. Open-source software for chemical reaction analysis using bond-electron matrices.
github.com/mrodobbe/Rxn-INSIGHT · 1k downloads per month

Solvent Explorer. Open-source chemical tool to analyze real-world solvent usage for different reaction classes.
www.solventexplorer.com · 2k visits per month

Research projects

PFAROS, Innovative Health Initiative 2026–present
Co-investigator
Co-supervisor of one PhD candidate. Five-year consortium; total project budget €48 million.

SMART-N, FWO postdoctoral project 2025–present
Principal investigator (personal fellowship) · Suspended until Aug 2027
Discovering Sustainable Molecules for Agriculture with Robotic Technology: Nitrification Inhibitor Design Using Artificial Intelligence and Self-Driving Labs. Three-year project.

UGent BOF project 2023–present
Co-investigator
Rational design and synthesis of non-hallucinogenic psilocin analogs as selective 5-HT$_{2A}$ receptor agonists with therapeutic potential. Four-year project.

TransPharm, Horizon Europe 2022–2026
Joint work package leader and co-investigator
Leading collaboration with Leuphana University, the University of York and Radboud University. Four-year consortium; total project budget €8 million.

FWO doctoral project 2021–2025
Principal investigator (personal fellowship)
Autonomous Drug Discovery: Machine Learning for Optimization of Computer-Aided Organic Synthesis in Flow. Four-year project.

Invited talks

What AI learns from a million reactions about solvent sustainability. 10th Green & Sustainable Chemistry Conference, Dresden, Germany (keynote). May 2026

Greener Pharmaceutical Production with a Digital Chemist. CESPE Conference, Ghent, Belgium. Oct 2025

Digitalising Chemical Reaction Engineering: From Candidate Molecules to Sustainable Production Processes. Department of Chemistry, University of York, UK. Jun 2025

AI-Driven Compound Screening and Reaction Planning for Greener Organic Synthesis. Advancing Green Chemistry Technologies Workshop, Ghent, Belgium. Dec 2024

Selected conference contributions (first author)

Assessing the Greenness of Organic Solvents Using Molecular Machine Learning. International Symposium on Green Chemistry (ISGC 2025), La Rochelle, France. May 2025

Artificial Intelligence for Reaction Planning in Continuous-Flow Reactors. International Symposium on Green Chemistry (ISGC 2025), La Rochelle, France. May 2025

Property Estimations with Chemical Accuracy for Industrially Relevant Molecules Using Geometric Deep Learning. ISCRE 28, Turku, Finland. Jun 2024

Ultrafast Solvation Property Prediction with Geometric Message Passing Neural Networks. ACS Fall Meeting, San Francisco, USA. Aug 2023

Reaction Data in Flow Chemistry: Power or Weakness? ACS Fall Meeting, San Francisco, USA. Aug 2023

Towards Big Data in Flow Chemistry: Popularity Trends and Best Practices. Flow Chemistry European Summit, Rotterdam, the Netherlands. Mar 2023

On the Relevance of Molecular Machine Learning in Industrial Chemistry. CRF-ChemCYS, Blankenberge, Belgium. Oct 2022

Teaching

Engineering Project 2021–2026
BSc in Chemical Engineering and Materials Science, Ghent University
Created a project module on AI in chemical engineering; taught four semesters of 12 weeks, with approximately 30 students per year. Combined lectures with active learning and organised company visits to BASF Antwerp.

Python in Chemical Reaction Engineering Sep 2022 – Apr 2023
Doctoral School in Engineering, Ghent University
Taught introductory Python in an accredited doctoral course for PhD researchers.

Supervision and mentoring

Doctoral co-supervision (co-promotor), Ghent University

PhD candidate	Co-supervisor	Period
Frances Aelbrecht	Prof. Kevin M. Van Geem; PFAROS project	2026–present
Senne Van de Velde	Prof. Kevin M. Van Geem	2026–present
Milan Bekaert	Prof. Bruno Linclau, Department of Chemistry	2026–present
Jiaying Wang	Prof. Kevin M. Van Geem	2025–present

PhD graduate	Thesis	Defended
Lowie Tomme	*Accelerated Generation of Detailed Chemical Reaction Networks*. Co-supervised with Prof. Kevin M. Van Geem and Prof. Christian V. Stevens, 2025–2026.	6 Oct 2026

Master's thesis mentoring, Ghent University

Student	Thesis	Year
Frances Aelbrecht	*Decoding Smell through Molecular Machine Learning*	2025–2026
Senne Van de Velde	*Multimodal Molecular Machine Learning for Solubility Prediction*	2025–2026
Kasper Huysentruyt	*Agentic AI for Chemical Information Extraction from Scientific Publications*	2025–2026
Menno Michiels	*Feasibility Study of Chlorosulfonic Acid Synthesis in Microreactors using Computational Fluid Dynamics*	2025–2026
Mauritz De Wilde	*Automated estimation of kinetic parameters for organic syntheses in continuous-flow reactors*	2024–2025
Gaston Dejaeghere	*Scoring the Sustainability of Organic Solvents with Machine Learning*	2023–2024
Florian Logghe	*Predicting Physical Properties of Ionic Liquids with Machine Learning*	2022–2023
Lowie Tomme	*Chemical Reaction Representation Learning for Activation Energy Prediction*	2021–2022
Yannick Ureel	*Active Machine Learning for Hydrocracking Process Conditions*	2020–2021

Curriculum development and educational service

Secretary, Programme Committee in Chemical Engineering and Materials Science 2020–2026

Ad hoc member, Curriculum Commission, Faculty of Engineering and Architecture 2020–2026

Academic service

Session chair, 10th Green & Sustainable Chemistry Conference, Dresden, Germany May 2026

Session chair, *Artificial intelligence applications in chemical engineering*, EFCE Artificial Intelligence in Chemical Engineering Symposium, Paris Dec 2024

Reviewer for Chemical Engineering Journal; Current Opinion in Chemical Engineering; Fuel; Journal of Chemical Information and Modeling; Chemical Engineering Science; Reaction Chemistry & Engineering; Journal of Physical Chemistry A; Communications Chemistry; Journal of Cheminformatics; Chemical Science; npj Computational Materials

Industrial experience

Research intern in computational chemistry Apr 2022 – Jul 2022
SABIC T&I, Geleen, the Netherlands, and Sugar Land, Texas, USA
Machine learning models for screening liquid-phase properties.

Intern in chemical process design Jul 2019 – Sep 2019
Yokogawa, Amersfoort, the Netherlands
Dynamic process simulation in Petro-SIM of an LNG calibration facility.

Intern in process optimisation Jan 2019 – Feb 2019
TE Connectivity, Oostkamp, Belgium
Data- and experiment-driven optimisation of more than 100 injection-moulding machines to enable recycled-plastic feedstocks.

Science communication

Flemish PhD Cup: “Nieuwe geneesmiddelen ontdekken met een AI-receptenboek” 2026

Eos Wetenschap: “Artificiële intelligentie behaalt diploma chemie” 2026

Coverage of the solvent study in *Nature*, *C&EN*, *Spektrum der Wissenschaft* 2026

Interview, *Spektrum der Wissenschaft*: “KI hat meine Erwartungen sogar übertroffen” 2025

Flemish Science Festival: “Is de chemicus van de toekomst een robot?” 2025

Languages

Dutch (native), English (C2), French (B2), German (B2).